

Arterial Blood Gas (การวัดแรงดันก๊าซในเลือดแดง)

	Acid	Base
pH		
PCO_2		
HCO_3		
PaO_2		
Compensate		

$$1. \text{ pH} = 7.28 \quad \text{PaCO}_2 \text{ (mmHg)} = 55 \quad \text{HCO}_3^- \text{ (mEq/L)} = 24$$

Respiratory Acidosis

	Acid	Base
pH	✓	
PCO ₂	✓	
HCO ₃		
PaO ₂		
Compensate	Uncompensate	

$$2. \text{ pH} = 7.52 \quad \text{PaCO}_2 \text{ (mmHg)} = 30 \quad \text{HCO}_3^- \text{ (mEq/L)} = 24$$

Respiratory Alkalosis

	Acid	Base
pH		✓
PCO ₂		✓
HCO ₃		
PaO ₂		
Compensate	Uncompensate	

$$3. \text{ pH} = 7.30 \quad \text{PaCO}_2 \text{ (mmHg)} = 55 \quad \text{HCO}_3^- \text{ (mEq/L)} = 30$$

Respiratory acidosis

	Acid	Base
pH	✓	
PCO ₂	✓	
HCO ₃		✓
PaO ₂		
Compensate	Partially compensate	

9. PH is not a base

148. Alkalosis

ต่อมา ถูว่าน้ำไทน์ก็ขึ้นได้ด้วยกับ PH

तथा $PACO_2$ = Respiratory

HCO_3^- : Metabolic

Compensate : Uncompensate = PH an 2 ตัว
ไม่เปลี่ยนค่า
Partial com " " = PH an
has 2 ตัว มี
เปลี่ยนค่า 1 ตัว
ไม่เปลี่ยนค่า 1 ตัว
PH
Fully compensate : PH an 1
ตัว 1 ตัว เปลี่ยนค่า

4. pH = 7.48 PaCO₂ (mmHg) = 30 HCO₃⁻ (mEq/L) = 20

	Acid	Base
pH		✓
PCO_2		✓
HCO_3	✓	
PaO_2		-
Compensate	Partially compensate	

Respiratory alkalosis

5. pH = 7.28 PaCO₂ (mmHg) = 30 HCO₃⁻ (mEq/L) = 15

	Acid	Base
pH	✓	
PCO_2		✓
HCO_3	✓	
PaO_2	-	
Compensate	Metabolic acidosis with partial compensation	

6. pH = 7.48 PaCO₂ (mmHg) = 48 HCO₃⁻ (mEq/L) = 32

	Acid	Base
pH		✓
PCO_2	✓	
HCO_3		✓
PaO_2	-	
Compensate	Metabolic alkalosis w. partial compensation	

7. pH = 7.36 PaCO₂ (mmHg) = 55 HCO₃⁻ (mEq/L) = 30

	Acid	Base
pH		
PCO ₂		
HCO ₃		
PaO ₂		
Compensate		

8. pH = 7.38 PaCO₂ (mmHg) = 30 HCO₃⁻ (mEq/L) = 18

	Acid	Base
pH		-
PCO ₂		✓
HCO ₃	✓	
PaO ₂	-	
Compensate	fully compensate	

9. pH = 7.15 PaCO₂ (mmHg) = 60 HCO₃⁻ (mEq/L) = 15

	Acid	Base
pH		
PCO ₂		
HCO ₃		
PaO ₂		
Compensate		

10. pH = 7.60 PaCO₂ (mmHg) = 25 HCO₃⁻ (mEq/L) = 34

	Acid	Base
pH		
PCO ₂		
HCO ₃		
PaO ₂		
Compensate		

Na⁺ = 135 - 145

K⁺ = 3.5 - 5

Ca²⁺ = 8.5 - 10.5

Calcium ion = 1.12 - 1.32

Magnesium = 1.6 (น้อยกว่า Hypo)

phosphate = 2.5 - 4.5 (0.81 - 1.45)

หัตถ์ในเลือด